

# E-commerce & Finance Insights Report

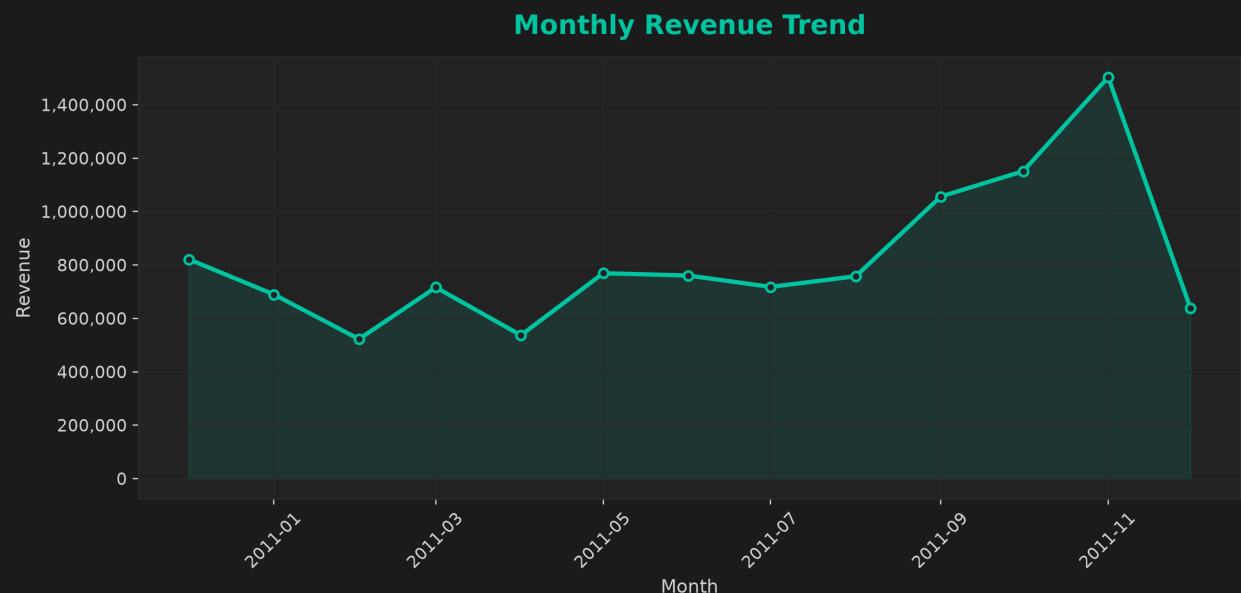
Generated via Python ReportLab • Author: Huzeif Khan

## Key Performance Indicators (KPIs)

- **Total Revenue:** 10,641,558.95
- **Total Customers:** 4,338
- **Average Order Value:** 2,453.10

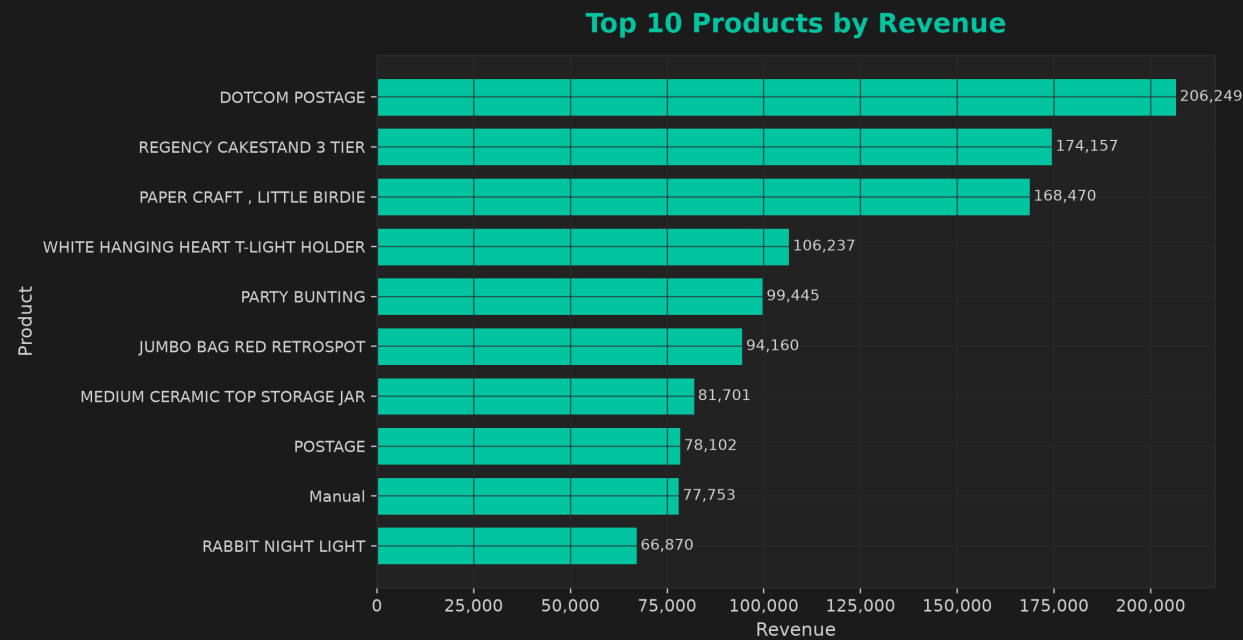
## Visual Summary

### Monthly Revenue Trend

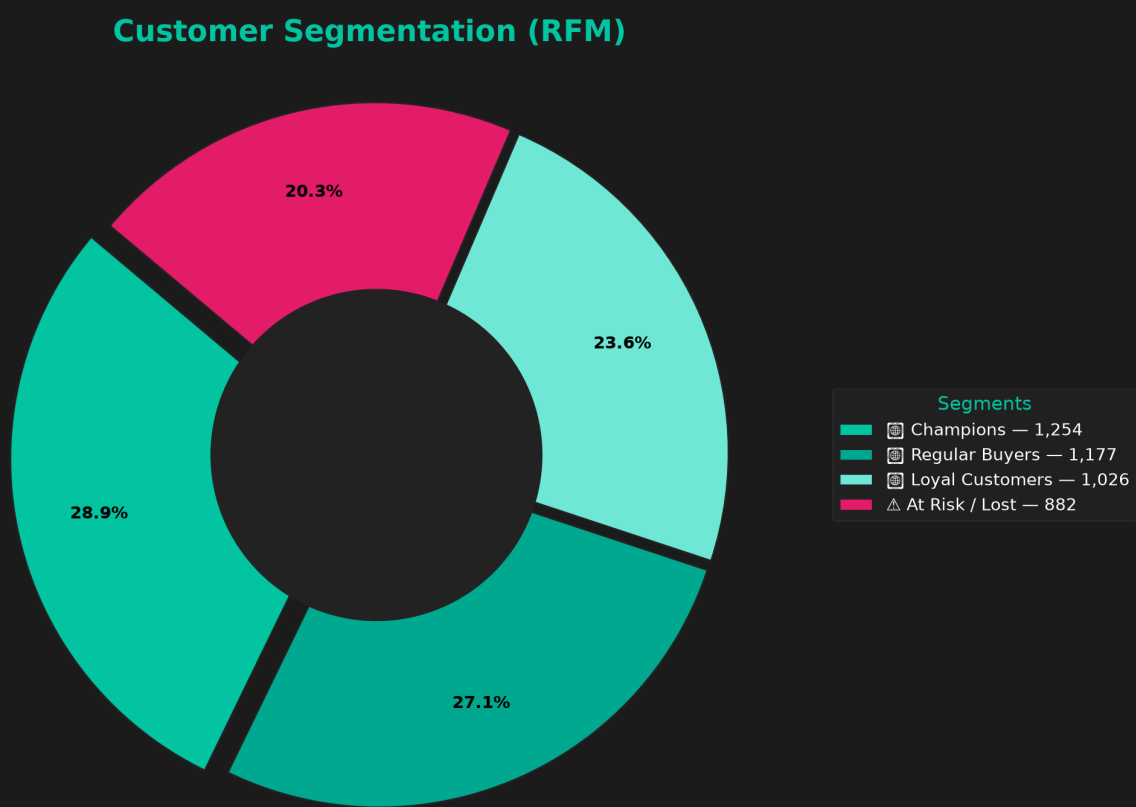


# Top 10 Products by Revenue

## Top 10 Products by Revenue



# Customer Segmentation (RFM Model)



# Cohort Retention

Each cell shows the % of customers who returned after N months, grouped by their first purchase month (cohort).



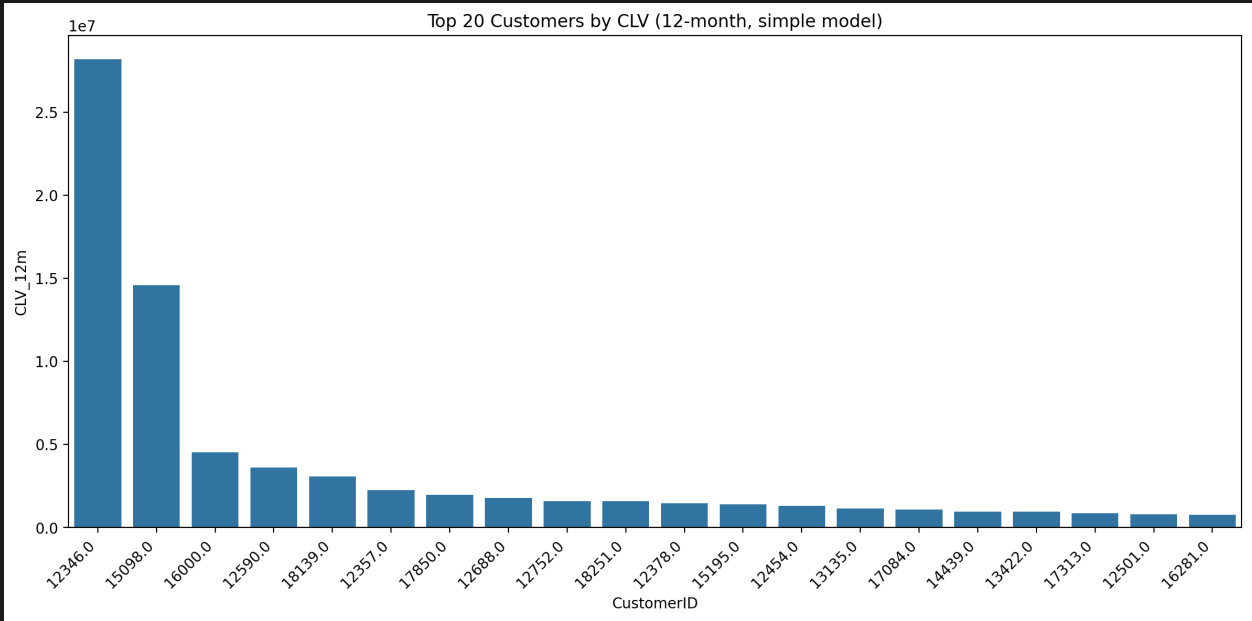
## Revenue Forecast (Next 12 Months)

SARIMAX forecast with 80% confidence band. Use this to plan inventory, marketing budgets, and hiring.



## Customer Lifetime Value (CLV) - 6-Month Outlook

We estimate a simple 6-month CLV as  $AOV \times (\text{avg monthly repeat probability}) \times 6$ , added to the historical revenue. Top 20 customers shown.



## Customer Lifetime Value (CLV v1 - 12-Month Model)

Deterministic CLV v1:  $AOV \times \text{Purchase Frequency per Month} \times 12 \text{ Months}$ . Shows your top-value customers based on average order value and repeat purchase rate.



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# CLV by Customer Segment

Average predicted 12-month CLV for each RFM segment.





## RFM x CLV - Segment Insights

How lifetime value varies by customer segment, and how it correlates with Recency, Frequency and Monetary (RFM).

- **Regular Buyers** - avg CLV  $\hat{a}$ , -125,288 (n=1177)
- **At Risk / Lost** - avg CLV  $\hat{a}$ , -81,282 (n=882)
- **Loyal Customers** - avg CLV  $\hat{a}$ , -43,711 (n=1026)
- **Champions** - avg CLV  $\hat{a}$ , -14,341 (n=1253)

### RFM-CLV Correlation



### CLV vs RFM (example scatter)



# Predictive CLV Modelling (Machine Learning)

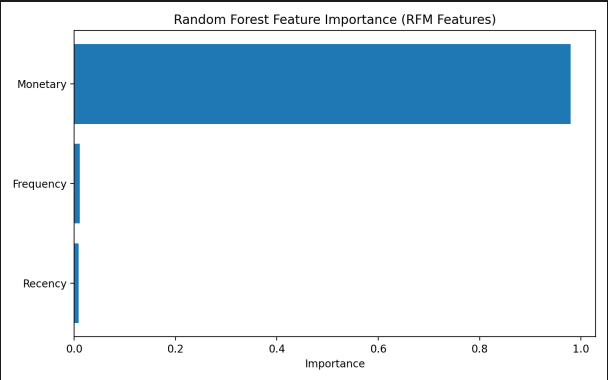
## Model Performance (RFM-Based CLV Prediction)

### Linear Regression

MAE: 0.00  
RMSE: 0.00  
R²: 1.00

### Random Forest Regressor

MAE: 43.66  
RMSE: 1052.86  
R²: 0.987



## Key Insights

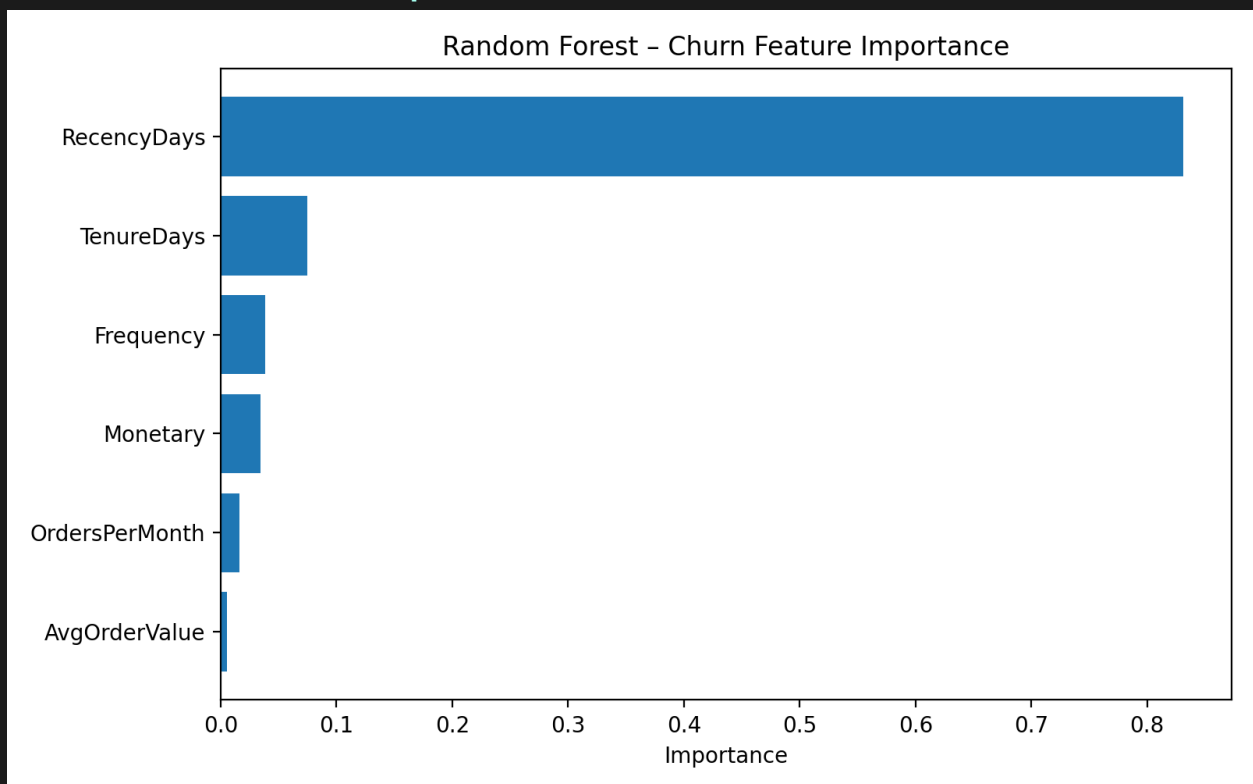
- RFM features (Recency, Frequency, Monetary) are exceptionally strong predictors of Customer Lifetime Value.
- Linear Regression achieved a perfect  $R^2 = 1.00$  due to the deterministic relationship between Monetary value and CLV.
- Random Forest ( $R^2 = 0.987$ ) confirms model stability and captures non-linear customer behaviours.
- Recency and Monetary value are the most influential features, reinforcing the validity of the RFM framework.
- Predictive modelling enhances segmentation accuracy and supports proactive customer value strategies.

## Customer Churn Prediction (Phase 10)

This section introduces a binary classification model predicting whether a customer is likely to become inactive based on purchase behavior. Churn is defined as no purchases in the last 90 days.

| Model                  | Accuracy | Precision | Recall | F1    | ROC-AUC |
|------------------------|----------|-----------|--------|-------|---------|
| LogisticRegression     | 0.992    | 1.000     | 0.975  | 0.987 | 1.000   |
| RandomForestClassifier | 1.000    | 1.000     | 1.000  | 1.000 | 1.000   |

### Random Forest Feature Importance



### Random Forest ROC Curve

